

## FARIBAULT, MN, USA

WMO: 726563

Lat: 44.333N

Lon: 93.317W

Elev: 1060

StdP: 14.14

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94969

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -13.5      | -8.5 | -20.2                         | 1.9 | -11.6 | -16.7 | 2.3 | -8.1 | 24.8                  | 15.8 | 22.5 | 15.2 | 4.9                   | 290  | 0.553 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.0                   | 90.1            | 74.2 | 87.6 | 72.8 | 83.9 | 70.8 | 77.7                | 86.4 | 75.5 | 83.2 | 73.6 | 81.0 | 10.4                 | 190  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|----------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)            |
| 74.9                            | 136.0 | 83.3 | 72.8 | 126.4 | 80.8 | 71.7 | 121.8 | 79.5 | 42.0          | 86.0 | 39.7 | 83.6 | 37.9 | 81.3 | 85.1           |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |  |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|--|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |  |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |  |
| 22.8              | 19.8 | 17.8 | DB                         | -20.3 | 94.9               | 8.1 | 2.6                                                | -26.1 | 96.7       | -30.9 | 98.3       | -35.4 | 99.7       | -41.3 | 101.6 |  |
|                   |      |      | WB                         | -20.3 | 81.2               | 7.7 | 2.4                                                | -25.8 | 82.9       | -30.3 | 84.2       | -34.6 | 85.6       | -40.2 | 87.3  |  |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 45.8   | 14.9  | 19.0  | 32.5  | 46.7  | 58.5  | 68.4  | 72.1  | 69.3  | 62.2  | 48.8  | 34.6  | 20.9  |
|                                                                   | DBStd       | 22.31  | 13.23 | 12.72 | 12.49 | 9.97  | 8.86  | 6.57  | 5.53  | 5.48  | 8.51  | 9.55  | 10.78 | 11.99 |
|                                                                   | HDD50       | 4231   | 1088  | 867   | 556   | 175   | 23    | 0     | 0     | 0     | 8     | 139   | 472   | 903   |
|                                                                   | HDD65       | 7648   | 1553  | 1287  | 1007  | 552   | 242   | 40    | 8     | 21    | 151   | 508   | 911   | 1368  |
|                                                                   | CDD50       | 2701   | 0     | 0     | 15    | 77    | 286   | 551   | 685   | 597   | 376   | 103   | 11    | 0     |
|                                                                   | CDD65       | 640    | 0     | 0     | 1     | 4     | 39    | 140   | 228   | 153   | 68    | 7     | 0     | 0     |
|                                                                   | CDH74       | 5868   | 0     | 0     | 3     | 60    | 460   | 1263  | 2150  | 1231  | 627   | 73    | 1     | 0     |
|                                                                   | CDH80       | 1768   | 0     | 0     | 0     | 16    | 153   | 395   | 715   | 305   | 171   | 13    | 0     | 0     |
|                                                                   | WSAvg       | 7.7    | 8.2   | 8.3   | 8.6   | 9.3   | 8.2   | 7.0   | 5.9   | 5.4   | 6.8   | 8.2   | 8.4   | 7.8   |
|                                                                   | PrecAvg     | 33.7   | 1.0   | 0.8   | 2.0   | 2.8   | 4.1   | 5.0   | 4.4   | 4.9   | 3.2   | 2.5   | 1.7   | 1.1   |
| Precipitation                                                     | PrecMax     | 40.8   | 2.7   | 1.9   | 3.8   | 5.6   | 8.1   | 9.1   | 8.8   | 10.3  | 10.0  | 5.8   | 5.1   | 3.7   |
|                                                                   | PrecMin     | 22.9   | 0.2   | 0.1   | 0.3   | 0.6   | 1.4   | 2.2   | 1.4   | 1.5   | 0.7   | 0.5   | 0.1   | 0.2   |
|                                                                   | PrecStd     | 4.6    | 0.6   | 0.5   | 0.9   | 1.3   | 1.9   | 1.8   | 2.0   | 2.0   | 2.1   | 1.5   | 1.3   | 0.7   |
|                                                                   |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 44.9  | 50.3  | 71.6  | 81.4  | 90.6  | 91.4  | 93.2  | 90.3  | 89.9  | 81.9  | 68.1  | 53.9  |
|                                                                   |             | MCWB   | 39.2  | 45.2  | 57.7  | 62.1  | 67.8  | 73.9  | 78.0  | 76.4  | 72.3  | 63.5  | 55.5  | 49.9  |
|                                                                   | 2%          | DB     | 38.7  | 44.7  | 62.8  | 73.0  | 83.9  | 88.4  | 90.3  | 86.2  | 84.2  | 74.8  | 61.2  | 45.2  |
|                                                                   |             | MCWB   | 35.1  | 40.7  | 54.3  | 56.9  | 64.6  | 71.0  | 76.5  | 74.5  | 70.3  | 60.7  | 51.9  | 41.4  |
|                                                                   | 5%          | DB     | 36.0  | 39.2  | 55.4  | 68.3  | 79.3  | 84.2  | 87.7  | 82.5  | 81.3  | 70.0  | 54.9  | 39.2  |
|                                                                   |             | MCWB   | 33.6  | 35.8  | 49.1  | 54.0  | 62.2  | 69.5  | 74.8  | 71.4  | 68.0  | 58.7  | 48.2  | 36.0  |
|                                                                   | 10%         | DB     | 32.4  | 36.4  | 50.1  | 63.5  | 73.3  | 81.4  | 83.8  | 81.2  | 77.2  | 64.0  | 49.8  | 36.4  |
|                                                                   |             | MCWB   | 31.2  | 33.9  | 45.0  | 51.6  | 59.2  | 67.5  | 72.5  | 70.4  | 65.7  | 55.4  | 44.1  | 34.2  |
|                                                                   | 0.4%        | WB     | 39.4  | 45.8  | 60.4  | 64.6  | 71.9  | 78.1  | 81.7  | 79.2  | 76.3  | 68.4  | 58.9  | 49.5  |
|                                                                   |             | MCDB   | 43.7  | 49.2  | 67.1  | 75.3  | 84.2  | 89.3  | 90.1  | 87.3  | 84.8  | 76.7  | 65.8  | 52.3  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 35.9  | 40.7  | 55.5  | 59.7  | 68.0  | 75.0  | 78.4  | 76.4  | 73.3  | 63.5  | 53.5  | 41.1  |
|                                                                   |             | MCDB   | 38.4  | 44.3  | 60.0  | 70.3  | 78.7  | 83.4  | 87.0  | 83.4  | 81.4  | 69.6  | 59.0  | 45.1  |
|                                                                   | 5%          | WB     | 33.4  | 36.7  | 50.4  | 56.3  | 65.3  | 72.4  | 76.3  | 74.3  | 70.6  | 60.0  | 48.9  | 37.0  |
|                                                                   |             | MCDB   | 35.6  | 39.4  | 56.0  | 65.4  | 74.6  | 81.2  | 84.5  | 80.8  | 77.7  | 67.3  | 54.8  | 39.4  |
|                                                                   | 10%         | WB     | 31.1  | 34.0  | 44.9  | 53.2  | 62.6  | 69.9  | 74.3  | 72.3  | 67.8  | 56.7  | 45.0  | 34.2  |
|                                                                   |             | MCDB   | 32.4  | 36.3  | 49.6  | 61.6  | 70.7  | 78.0  | 81.8  | 78.7  | 74.8  | 63.8  | 49.0  | 36.2  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 16.5  | 17.0  | 17.5  | 19.9  | 20.6  | 19.6  | 20.0  | 19.6  | 21.7  | 19.4  | 16.0  | 14.5  |
|                                                                   |             | MCDBR  | 18.0  | 17.3  | 23.7  | 28.4  | 28.0  | 24.0  | 22.6  | 21.5  | 24.8  | 26.8  | 23.9  | 17.9  |
|                                                                   |             | MCWBR  | 15.2  | 14.2  | 16.1  | 15.6  | 13.1  | 10.9  | 11.6  | 11.7  | 12.6  | 14.8  | 15.9  | 14.5  |
|                                                                   | 5% WB       | MCDBR  | 16.7  | 16.3  | 22.4  | 24.3  | 22.4  | 20.2  | 20.1  | 18.9  | 20.8  | 21.8  | 22.3  | 16.4  |
|                                                                   |             | MCWBR  | 14.5  | 13.5  | 16.0  | 15.3  | 12.1  | 11.6  | 11.8  | 11.9  | 12.5  | 14.1  | 15.7  | 13.7  |
|                                                                   |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.269 | 0.288 | 0.317 | 0.371 | 0.398 | 0.405 | 0.406 | 0.403 | 0.374 | 0.323 | 0.292 | 0.268 |
|                                                                   | taud        |        | 2.426 | 2.354 | 2.401 | 2.303 | 2.284 | 2.305 | 2.308 | 2.315 | 2.365 | 2.501 | 2.502 | 2.445 |
|                                                                   | Ebn at Noon |        | 275   | 287   | 292   | 283   | 277   | 274   | 272   | 270   | 269   | 272   | 264   | 264   |
|                                                                   | Edh at Noon |        | 23    | 30    | 32    | 39    | 41    | 40    | 40    | 38    | 34    | 26    | 21    | 20    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 555   | 899   | 1177  | 1450  | 1696  | 1896  | 2011  | 1683  | 1326  | 855   | 551   | 399   |
|                                                                   | RadStd      |        | 67    | 95    | 116   | 156   | 137   | 155   | 81    | 119   | 97    | 113   | 53    | 36    |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (60 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page